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Calculation

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Two exercises, in which you will solve the differential equation for the intensity of the light in a layer of sunscreen.

Exercise A

1/1 point (graded)

The boundary value problem for the UV light intensity in the sunscreen lotion layer is:

$$\frac{dI}{dx} = -kI(x), \quad I(0) = I_{full}.$$

The analytical solution is

☒ $I(x) = I_{full} e^{-kx}$ ✓

☐ $I(x) = e^{-kx} + I_{full} - 1$

☐ $I(x) = k e^{-x} + I_{full} - k$

☐ $I(x) = I_{full} e^{kx}$

Submit

You have used 1 of 1 attempt

i Answers are displayed within the problem

Exercise B

1/1 point (graded)

Sunscreen lotion with SPF=30 is calibrated such that when a $20\mu\text{m}$ layer has been applied, the UV intensity at the skin has been reduced to 1/30-th of the UV intensity of the sunlight.

Use a calculator and calculate the value of k in two decimals.

✓ Answer: 0.17

You have used 1 of 3 attempts

i Answers are displayed within the problem

In the next section, you will validate this calculation of the light intensity: how usefull are these results?

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